

DSA 210 - Final Project Report

Project Title: Multi-Factorial Analysis of NYC Airbnb Pricing

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1. Motivation

New York City is one of the most complex real estate markets globally. While traditional factors like property size define prices, urban dynamics such as safety and transportation accessibility are often claimed to be primary drivers of value. The motivation behind this project is to quantitatively test these claims. By integrating disparate datasets (Airbnb, Crime, and Metro), I aim to reveal how significantly public safety and transit density actually influence short-term rental costs, or if other engineered features like market segmentation play a more dominant role.

2. Data Source

To ensure a comprehensive analysis, I utilized three distinct datasets:

- **Airbnb Open Data (Kaggle):** Provides core information on listings, including price, neighborhood, and host details.
- **NYPD Crime Data (NYC Open Data):** Offers a historical record of criminal activities across the five boroughs, serving as a proxy for public safety.
- **MTA Subway Stations (NYC Open Data):** Contains precise geographical locations of subway entrances, used to measure transit accessibility.

3. Data Cleaning and Exploratory Data Analysis (EDA)

3.1 Data Cleaning

Raw data required significant preprocessing:

- **Currency Standardization:** Airbnb price strings (e.g., "\$1,200") were converted to float values.
- **Normalization:** Borough names across all three datasets were standardized to uppercase (e.g., "Brooklyn" to "BROOKLYN") to ensure seamless merging.
- **Missing Values:** Null entries in crime and metro counts were treated as zero to maintain dataset integrity.

3.2 Exploratory Data Analysis (EDA)

Through EDA, I identified that Airbnb prices are heavily skewed toward certain neighborhoods. A correlation analysis (Heatmap) revealed that while simple crime counts have a weak linear

correlation with price, the "Price Category" feature I engineered showed a much stronger relationship, suggesting that price tiers are more informative than raw dollar amounts.

4. Machine Learning Models

I employed a **Random Forest Regressor** to predict rental prices. This model was chosen for its ability to handle non-linear relationships and its robustness against outliers in the crime and price data.

- **Target Variable:** Price
- **Features:** Neighborhood Group, Total Crime, Metro Count, and Price Category.
- **Performance:** The model achieved an **R² score of 0.745** and a Mean Absolute Error (MAE) of approximately \$99, indicating high predictive accuracy for the NYC market.

5. Findings from Machine Learning Model

The model's **Feature Importance** analysis provided key insights:

- **Price Category:** The most significant predictor, confirming that segmenting the market into Room, Apartment, and Penthouse tiers captures the variance better than raw locations alone.
- **Metro Access:** Interestingly, metro count is a baseline factor; its high density across NYC means it doesn't differentiate luxury listings from budget ones as much as expected.
- **Safety vs. Luxury:** Premium listings (Penthouses) exist even in boroughs with higher total crime counts, suggesting that building-specific security or proximity to local hubs outweighs borough-wide crime statistics.

6. Conclusion

The project successfully integrated urban data to model Airbnb pricing. The findings suggest that while public safety and transit are important, they act more as baseline requirements in the NYC market. The high R² score demonstrates that engineered categories and borough-level data can effectively predict price, providing a solid foundation for further granular studies at the neighborhood or street level.